Statistical Analysis of Federal Interest Rate and Inflation Data

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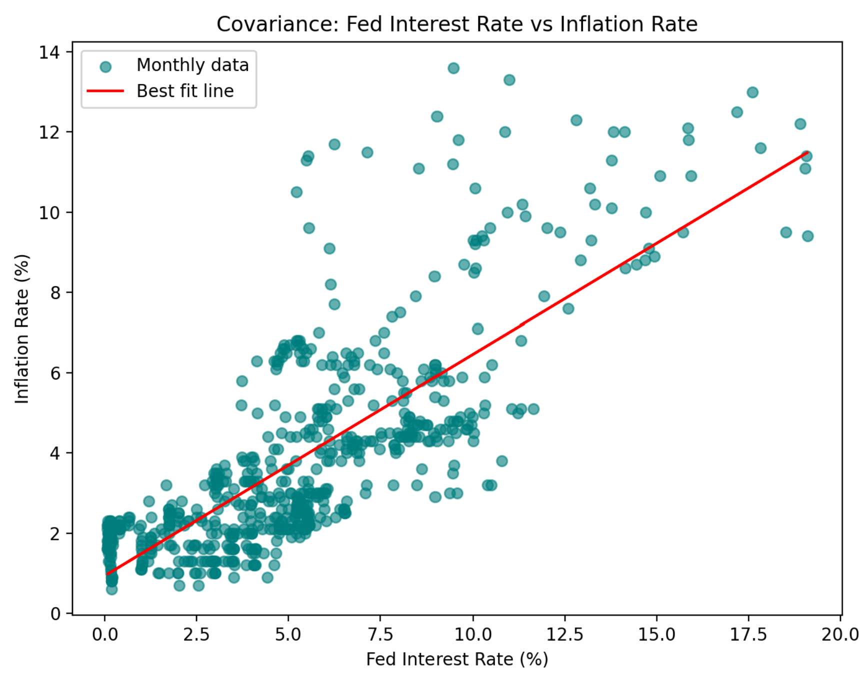
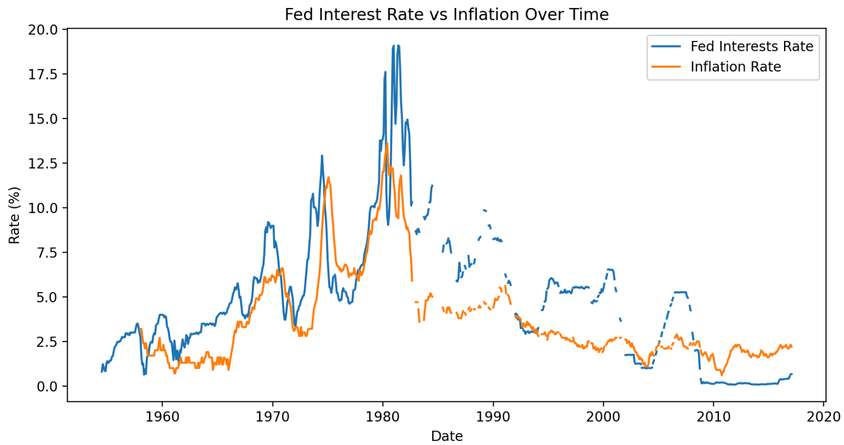
Summary:

In this project, I analyzed over 60 years of monthly Federal interest rate and inflation data using Python. I computed core statistics, mean, variance, and standard deviation, to quantify variability, and calculated covariance to measure the relationship between interest rates and inflation over time. Graphical tools, including line plots, rolling averages, lines of best fit, heat maps, and scatter plots, made these patterns easier to interpret by highlighting trends, periods of stability, and sudden deviations from expected behavior.

Methods / Findings:

Covariance between Federal interest rates and inflation was positive (7.378), indicating that the two variables generally moved together over time. This relationship is reinforced by both the scatter plot and its corresponding line of best fit, which exhibit a clear upward trend, as well as by the time series plots, where periods of rising and falling rates closely coincide.





<u>All subsequent analysis uses inflation data, which is more complete over the sample period and contains less null values.</u>

To capture how the data behaves over time I implemented a

<u>Markov chain model</u>: Each month was assigned a state based on inflation surprises, (categorized as negative(0), neutral(1), or positive(2), relative to recent trends).

```
state
1     555
2      81
0      74
```

The image above represents the observations per each regime on the last day of available data. This distribution of observations across regimes confirms that the neutral regime is the most frequent state. Although the 'Inflation Over Time' graph displays significant historical spikes, (most notably in the late 1970s), these represent relatively short lived periods of high volatility. When considering the full 60 year sample, the data exhibits high regime persistence, spending the vast majority of the time in a low variance and stable state.

From these regime observations a **transition matrix** is computed.

Transition Matrix:

	0	1	2
0	0.716216	0.270270	0.013514
1	0.036101	0.920578	0.043321
2	0.012346	0.296296	0.691358

(Transition matrix for last day of available data)

The corresponding transition matrix quantifies this persistence: the neutral regime persists month-to-month 92% of the time, negative shocks persist 72% of the time, and positive shocks persist 69% of the time (all conditional on the current state). Extreme regimes are less common than neutral, and when they do occur, tend to revert toward neutral rather than flip directly to the opposite extreme.

Regime statistics were used to quantify the mean and standard deviation of surprises/shocks within each state. It was observed that negative shocks averaged μ = **–0.94**(outcomes far below expectations), neutral months clustered near **0**(outcomes were close to expectations), and positive shocks averaged μ = **+0.88**(outcomes exceeded expectations).

Regime Statistics:

state	mean	std	count
0	-0.940631	0.514250	74
1	-0.008412	0.186233	554
2	0.881481	0.489753	81

Chebyshev's inequality was used to bound the frequency of large deviations within each regime. For k = 2, Chebyshev guarantees no more than 25% of observations fall beyond two standard deviations (a worst case bound that makes no assumption of normality). Observed violation rates were far lower: 5.4% in the negative-shock regime, 2.3% in the neutral regime, and 6.2% in the positive-shock regime. This gap between the theoretical bound and observed frequency confirms the regimes capture most of the variation in the data, and that the underlying distributions are better behaved than the worst case Chebyshev allows for.

```
Chebyshev Violation Rates:
 state
0    0.054054
1    0.023423
2    0.061728
```

Moment generating functions were calculated for each regime to give a much fuller description of their distributions. Because derivatives of the MGF at t = 0 yield the distribution's moments, MGFs capture not just mean and variance but skewness and kurtosis which enables estimation of tail probabilities conditional on the current regime.

States 0 and 2 illustrate this, while both have similar standard deviations and variance, their MGFs are mirror images. Because their MGFs differ substantially, this indicates asymmetries in higher-order moments and suggests different distributions of extreme outcomes.

	1	0	2
-1.0	1.025902	2.972455	0.455045
-0.9	1.021718	2.626950	0.488978
-0.8	1.017893	2.329601	0.526092
-0.7	1.014425	2.072924	0.566775
-0.6	1.011313	1.850675	0.611473
-0.5	1.008553	1.657636	0.660705
-0.4	1.006145	1.489438	0.715074
-0.3	1.004086	1.342417	0.775284
-0.2	1.002377	1.213496	0.842163
-0.1	1.001015	1.100085	0.916684
0.0	1.000000	1.000000	1.000000
0.1	0.999332	0.911395	1.093476
0.2	0.999011	0.832707	1.198742
0.3	0.999038	0.762612	1.317742
0.4	0.999412	0.699981	1.452815
0.5	1.000134	0.643855	1.606773
0.6	1.001206	0.593413	1.783016
0.7	1.002628	0.547953	1.985664
0.8	1.004403	0.506872	2.219731
0.9	1.006532	0.469650	2.491326
1.0	1.009017	0.435841	2.807927

Example: Consider an investor positioned for rising inflation while in state 2. The transition matrix shows state 2 is highly persistent, supporting confidence that the regime will hold. State 2's MGF shows rapid growth for $t > 0$, suggesting that positive deviations are influenced more by occasional large shocks than by a sequence of smaller increases. Chebyshev alone would only bound total deviation risk at 25% symmetrically; the MGF reveals that risk is skewed firmly toward the upside. Together, these tools suggest that upside deviations may dominate downside risk while the system remains in state 2, though further backtesting would be required to determine whether this information can be translated into profitable trading strategies.

Room for Improvement:

Applying these methods to live stock market data would extend this from descriptive regime analysis to a probability driven trading framework, using MGFs to characterize expected tail behavior and Markov chains to estimate regime persistence. Backtesting this framework against historical data would reveal its potential to generate profits.

Conclusion:

Overall, this analysis demonstrates how multiple statistical tools can be combined to provide a deeper understanding of the behavior of Federal interest rates and inflation over time. Summary statistics and visualizations highlight long term trends and periods of heightened volatility; Markov chain analysis captures the persistence of economic regimes and the likelihood of transitions between them; Chebyshev's inequality provides bounds on the frequency of extreme deviations, and moment generating functions offer additional insight into the distributional characteristics and risk profiles associated with each regime. Taken together, these methods transform historical data into a structured quantitative framework, allowing both common patterns and rare events to be analyzed in a measurable and interpretable manner.